distant origin travel along the fabric of the universe; if the fabric is warped, the motion of the photons will be affected much as for a material body. The bending of the path of light is greatest for those light signals that just graze the sun on their way to earth. A solar eclipse makes it possible to see such sun-grazing starlight without its being completely obscured by sunlight itself.

The angle through which the light path is bent can be measured in a simple way. The bending of the starlight's path results in a shift in the apparent position of the star. The shift can be accurately measured by comparing this apparent position with the star's actual location known from observations of the star at night (in the absence of the sun's warping influence), carried out when the earth is at an appropriate position, some six months earlier or later. In November of 1915, Einstein used his new understanding of gravity to calculate the angle through which starlight signals that just graze the sun would be bent and found the answer to be about .00049 of a degree (1.75 arcseconds, where an arcsecond is 1/3600 of a degree). This tiny angle is equal to that subtended by a quarter placed upright and viewed from nearly two miles away. The detection of such a small angle was, however, within reach of the technology of the day. At the urging of Sir Frank Dyson, director of the Greenwich observatory, Sir Arthur Eddington, a well-known astronomer and secretary of the Royal Astronomical Society in England, organized an expedition to the island of Principe off the coast of West Africa to test Einstein's prediction during the solar eclipse of May 29, 1919.

On November 6, 1919, after some five months of analysis of the photographs taken during the eclipse at Principe (and of other photographs of the eclipse taken by a second British team led by Charles Davidson and Andrew Crommelin in Sobral, Brazil), it was announced at a joint meeting of the Royal Society and the Royal Astronomical Society that Einstein's prediction based on general relativity had been confirmed. It took little time for word of this success—a complete overturning of previous conceptions of space and time—to spread well beyond the confines of the physics community, making Einstein a celebrated figure worldwide. On November 7, 1919, the headline in the London Times read "REVOLUTION IN SCIENCE